CSCI 5526

Machine Learning for Healthcare: History and Unique Challenges

[Content inspired from MIT Machine Learning for Healthcare class lecture 1]

Announcements

- Lab 0 was **due yesterday!**
- MP1 released today.
  - Due 9/25 at 11.59pm.
- Project:
  - ***REQUIRED*** group size: 3-4
  - Final project resources: https://yoga-varatha.github.io/ml4health-umn/finalproject/
  - Survey: https://z.umn.edu/ml4h_project_survey (until 9/20)

Exit pass responses

- Can we bring written notes to the quiz?
- What would be the targeted contents of the quizzes? Do we need to bring pens ourselves?

-- Format: in-person during class.

-- Questions: multiple choice or short answer questions.

-- Duration: 20-30 minutes

-- Topics: lectures and assignments covered before the quiz.

-- Open note: can bring notes, either printed or electronic.

- Are there any extra credit opportunities?

Not currently. Will consider if there is a legitimate reason.

Exit pass responses

- If machine learning is never used without oversight, why is deep learning so commonly used if it lacks interpretability? Is the performance boost high enough to warrant the loss of interpretability compared to simpler models?

-- Yes, essentially. For unstructured data (e.g., images), the performance boost is substantial.

-- You can have oversight without absolute transparency.

-- Rigorous validation

-- Explainable AI workarounds

-- Keep deep learning strictly for supportive tasks

Exit pass responses

- If an AI system highlights the regions that contributed to its prediction, could this create automation bias or anchoring by directing clinicians' attention toward those regions? How can we design explanations so that they help clinicians without making them over-trust potentially noisy or spurious features?

-- Interesting question

-- Some strategies:

    -- Provide contextual uncertainty metrics

    -- Introduce desynchronized or on-demand explanations

Exit pass responses

- What are the deployment method and expectations in this course?

-- We do not expect you to deploy anything in real clinical environments.

-- But we expect that you apply rigorous validation strategies.

-- Consider robustness, fairness, interpretability, etc.

Exit pass responses

- There has been a lot of talk about many people losing their jobs due to AI/ML. Do you think this will ever impact healthcare jobs, and if you do when do you think these impacts will be seen?

-- Some areas have seen impacts.

-- Medical Scribes and Dictation

-- Corporate and Insurance Adjustments

-- IT and informatics

Exit pass responses

- How the signals collected from Human Beings different from signals collected from animals? How can we work on those real experimental datasets?

-- They are very different. Will need specialized algorithms.

- Will you upload the PPT a day before class?

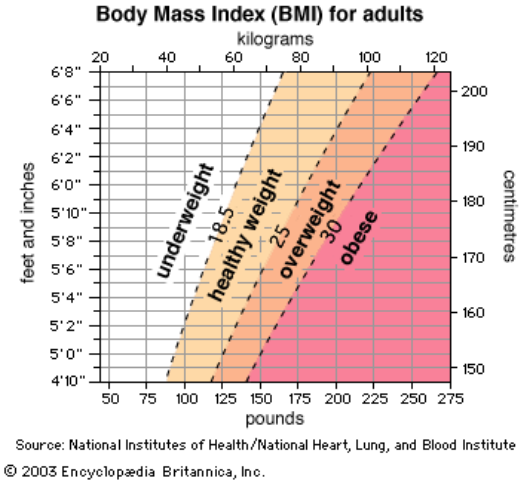
-- I will upload same day, before class.

Outline

- History of algorithms and AI in healthcare
- Historical challenges
- Why now? What changed?
  - Data, standardization
  - Computing and algorithmic advances
- What is different in healthcare ML?
  - Cost of errors
  - Data, labeling inconsistencies
  - Security and privacy
  - The need for trustworthiness (robust, explainable, fair, equitable, accountable)
- Some deployment experiences

Algorithms in Healthcare

- Human designed
  - BMI (body mass index) – 19th century
  - TNM (Tumor-Node-Metastasis) staging – 1953
  - PHQ-9 (patient health questionnaire) – 1999



- AI-based/expert systems
  - MYCIN – 1970s (identifies severe bacterial infections)
  - CASNET – 1970s (causal association network for diagnosing glaucoma)
  - INTERNIST-1 – 1980s (diagnosis for internal medicine)

MYCIN Expert System (1970s)

Stanford

- Expert system for identifying bacteria caused severe infections.
- Proposed a good therapy in 69% of the cases.
- Better than experts.

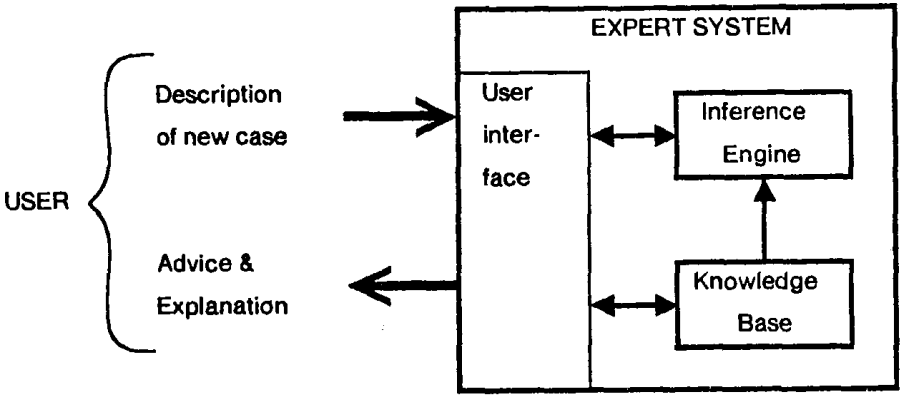


FIGURE 1-1 Major parts of an expert system. Arrows indicate information flow.

Dialogue interface

I am ready

** THIS IS A 26 YEAR OLD MALE PATIENT

My understanding is:
The age of the patient is 26
The sex of the patient is male

** FIVE DAYS AGO, HE HAD RESPIRATORY-TRACT SYMPTOMS

What is his name?

** JO

My understanding is:
The name of the patient is Jo
Respiratory-tract is one of the symptoms that the patient had

** A COUPLE OF DAYS BEFORE THE ADMISSION, HE HAD A MALAISE

Please give me the date of admission

** MARCH 12, 1979

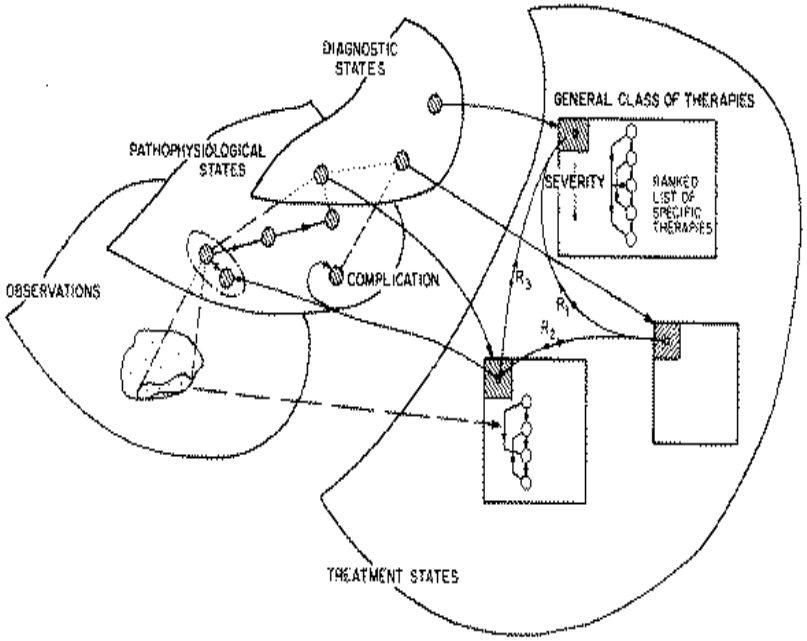
My understanding is:
The patient was admitted at the hospital 3 days ago
Malaise is one of the symptoms that the patient had 5 days ago

FIGURE 33-1 Short sample dialogue. The physician's inputs appear in capital letters after the double asterisks.

CASNET Consultation System (1970s)

Rutgers

- A causal-associational network (CASNET) model for describing disease processes.
- The system consists of four main elements:
  1. Observations or findings of the patient-signs, symptoms and test results.
  2. Pathophysiological states that describe internal abnormal conditions or mechanisms.
  3. Disease states that are defined at a higher level of abstraction.
  4. Treatment plans or general classes of therapies.

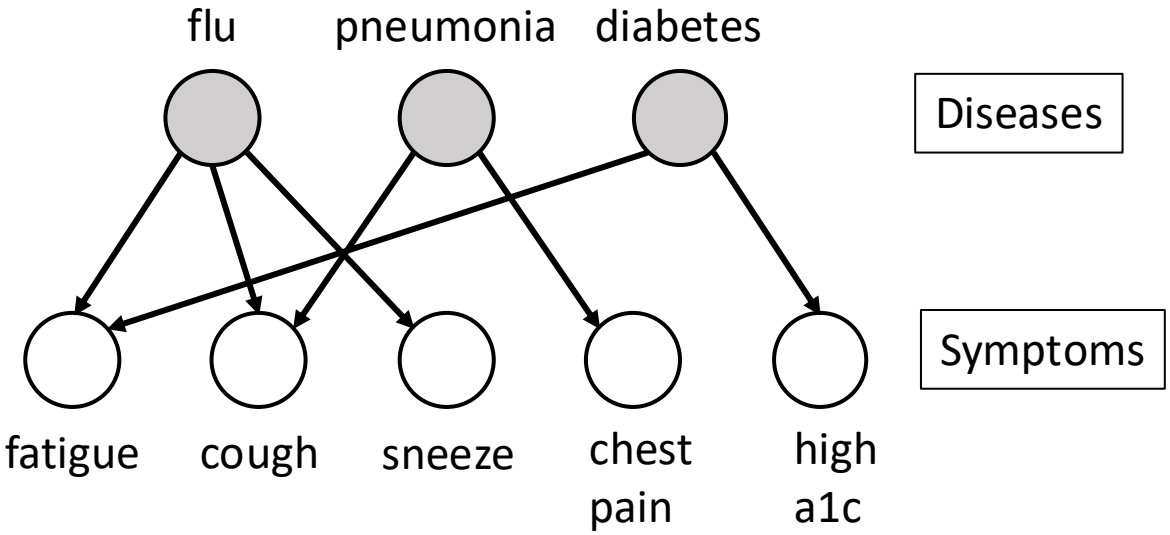


INTERNIST-1 Expert System (1980s)

U of Pittsburgh

- Provides diagnostic assistance in internal medicine.
- Probabilistic model:
  - 570 binary disease variables
  - 4,075 binary symptom variables
  - 45,470 directed edges

15 person-years of work!



Historical Challenges

- Data and labeling
  - Electronic records were not mainstream
  - Tons of manual effort
- Standardization
  - Lots of different archiving standards between hospitals
  - Minimal interoperability
- Computing and algorithms
  - Powerful computers were not available
  - Only rule-based algorithms

What changed?

Adoption of Electronic Health Records

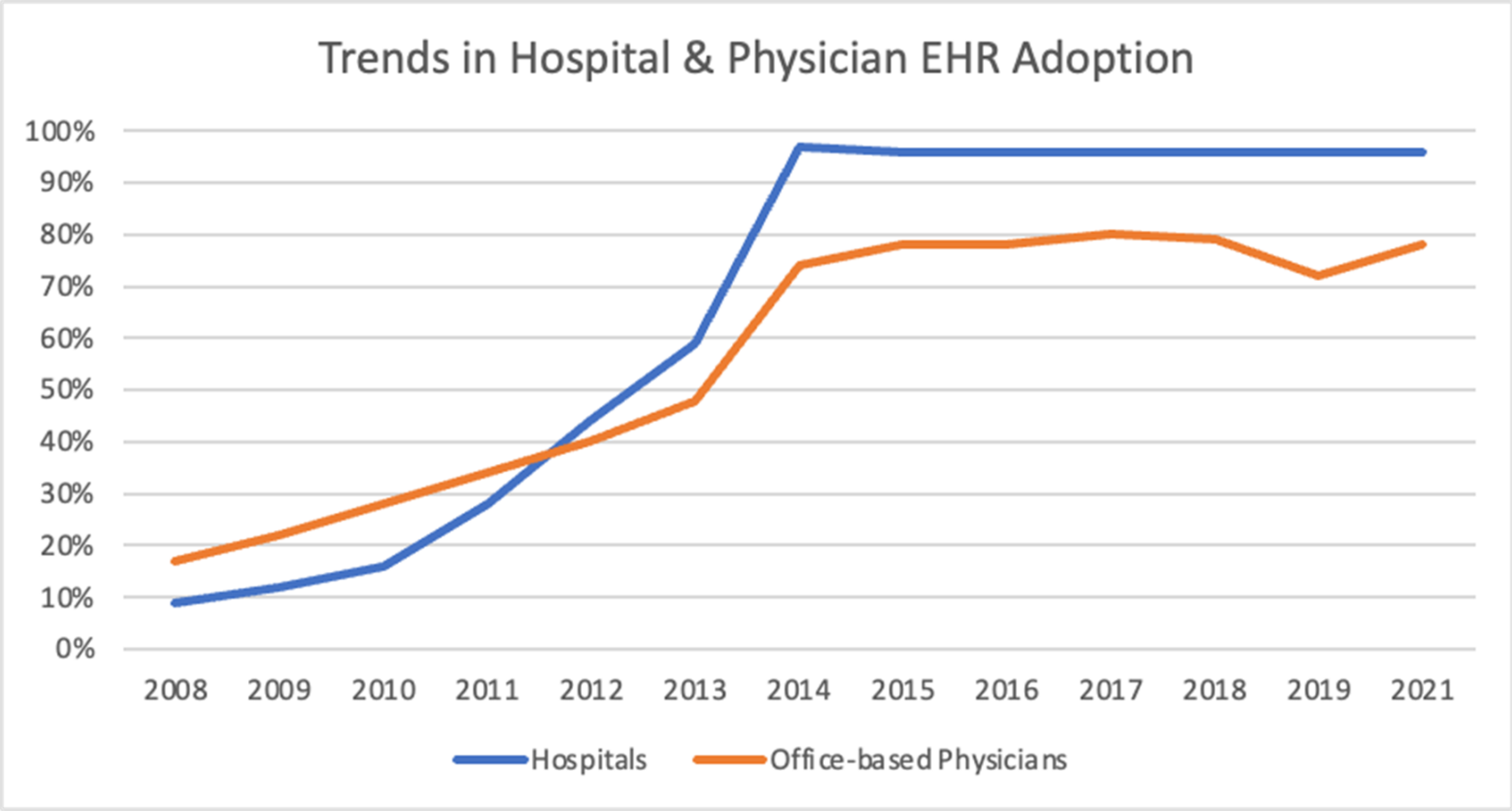
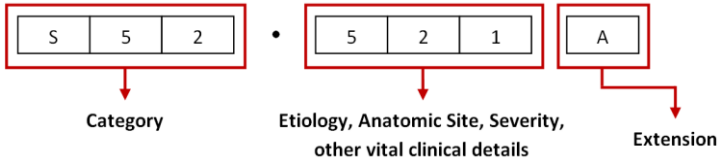


Image taken from healthit.gov.

Standardization

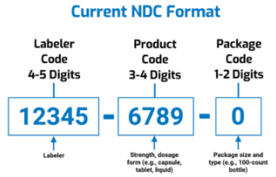
- ICD-9 and ICD-10 diagnosis codes



- Laboratory test orders and results (LIONC) codes

LOINC		ORRESU		STRNU	
LOINC	Text	NCICode	Text	NCICode	Text
8329-5	Core Body Temperature	C44277	F	C42559	C
8329-5	Core Body Temperature	C42559	C	C42559	C

- National drug codes (NDCs)



- Unified Medical Language System (UMLS)
- etc.

Large Open Datasets

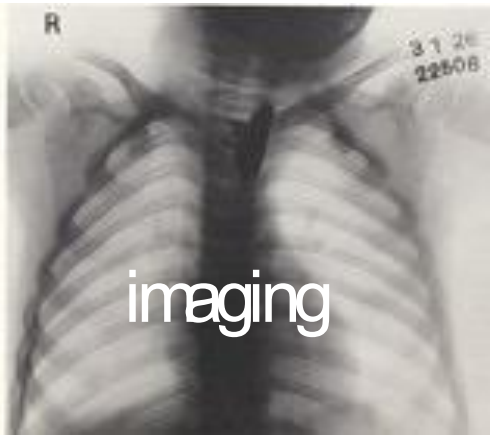
- MIMIC – PhysioNet
  - Deidentified data from ~40K critical care patients
- Multiple imaging datasets
  - Human Connectome Project
  - Alzheimer's disease neuroimaging initiative (ADNI)
  - Adolescent brain cognitive development study (ABCD)
- TCGA (The Cancer Genome Atlas Program)
- The UK Biobank
- All of Us database
- And many more…

Multiple Digital Data Sources



proteomics





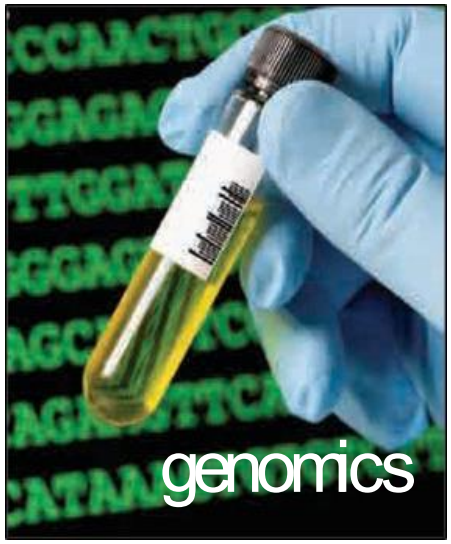




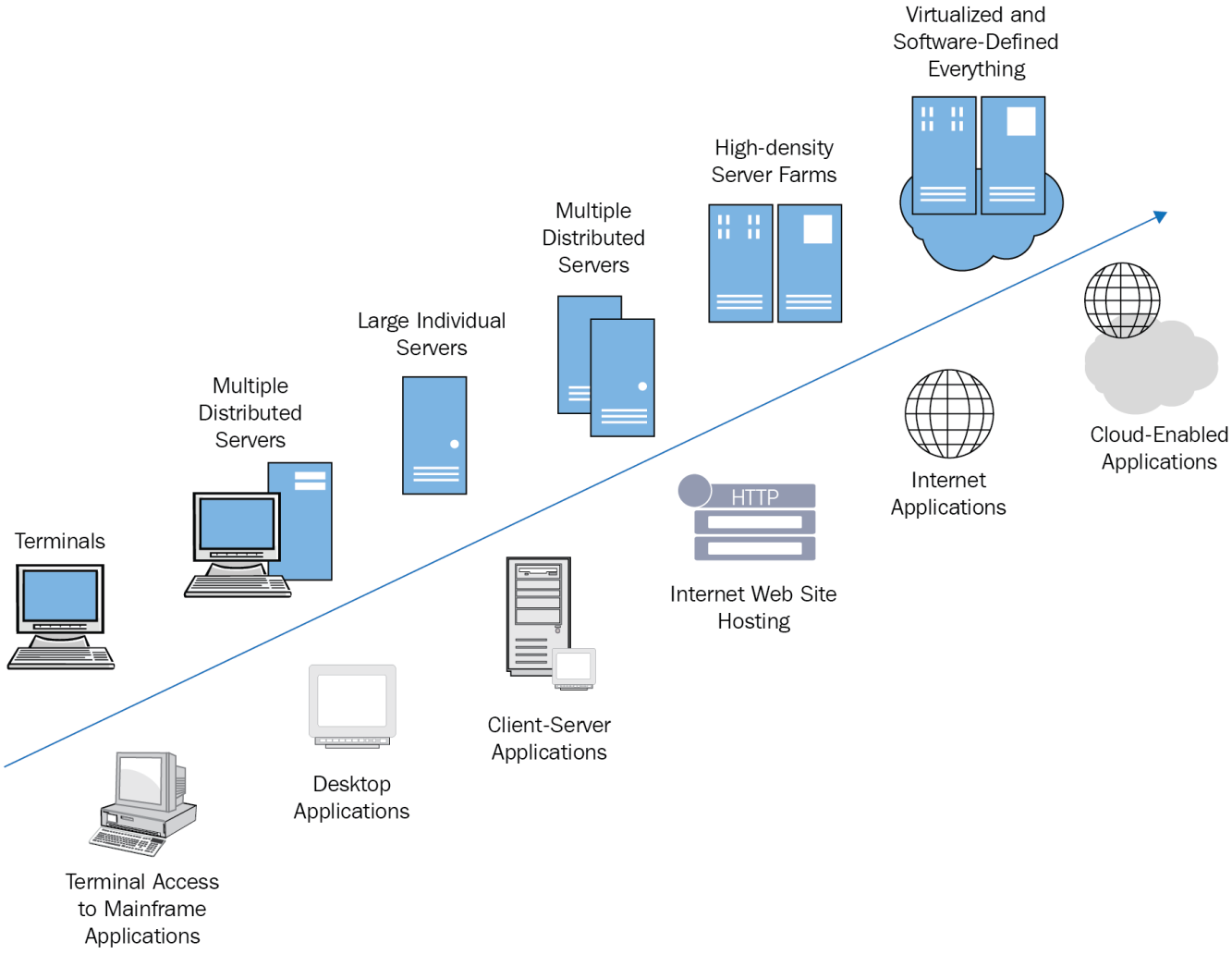
social media







Computing Advances

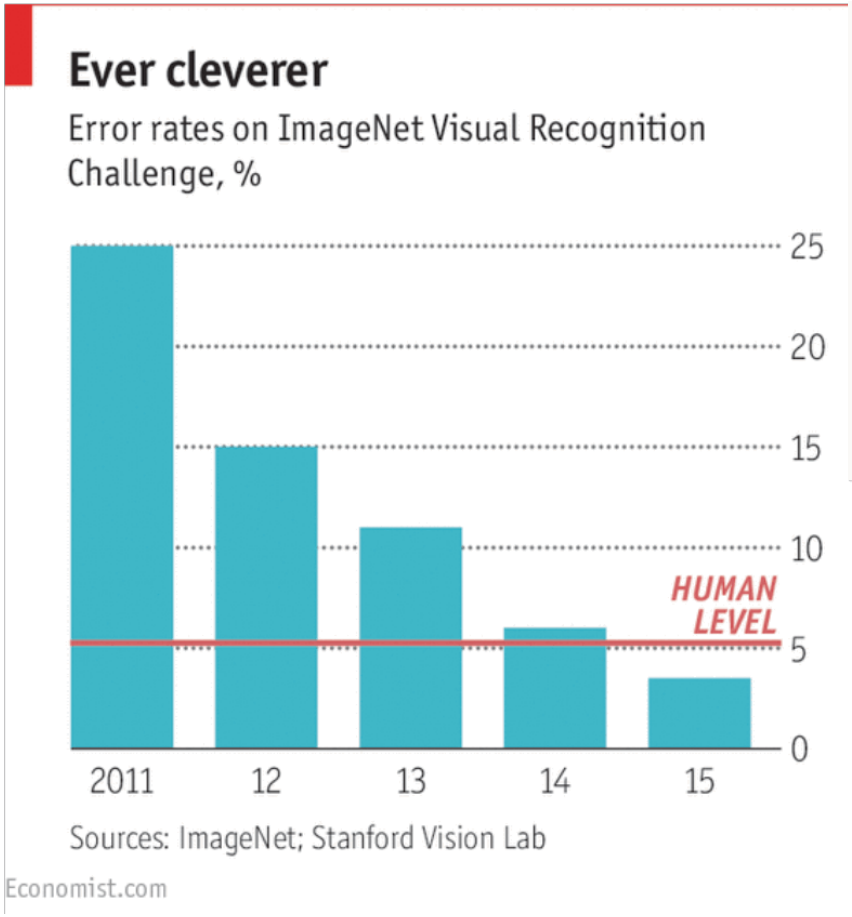


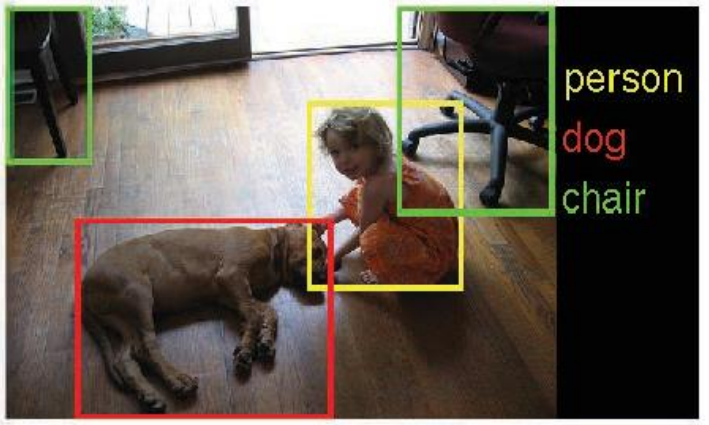
Algorithmic Advances

- Major advances in ML & AI
  - Learning with high-dimensional features (e.g., l1- regularization)
  - Semi-supervised and unsupervised learning
  - Modern deep learning techniques (e.g., convnets, transformers)

- Democratization of machine learning
  - High quality open-source software, such as Python's scikit-learn, TensorFlow, Torch
  - Pretrained models such as VGG, inception, ResNet, llama

Algorithmic Advances

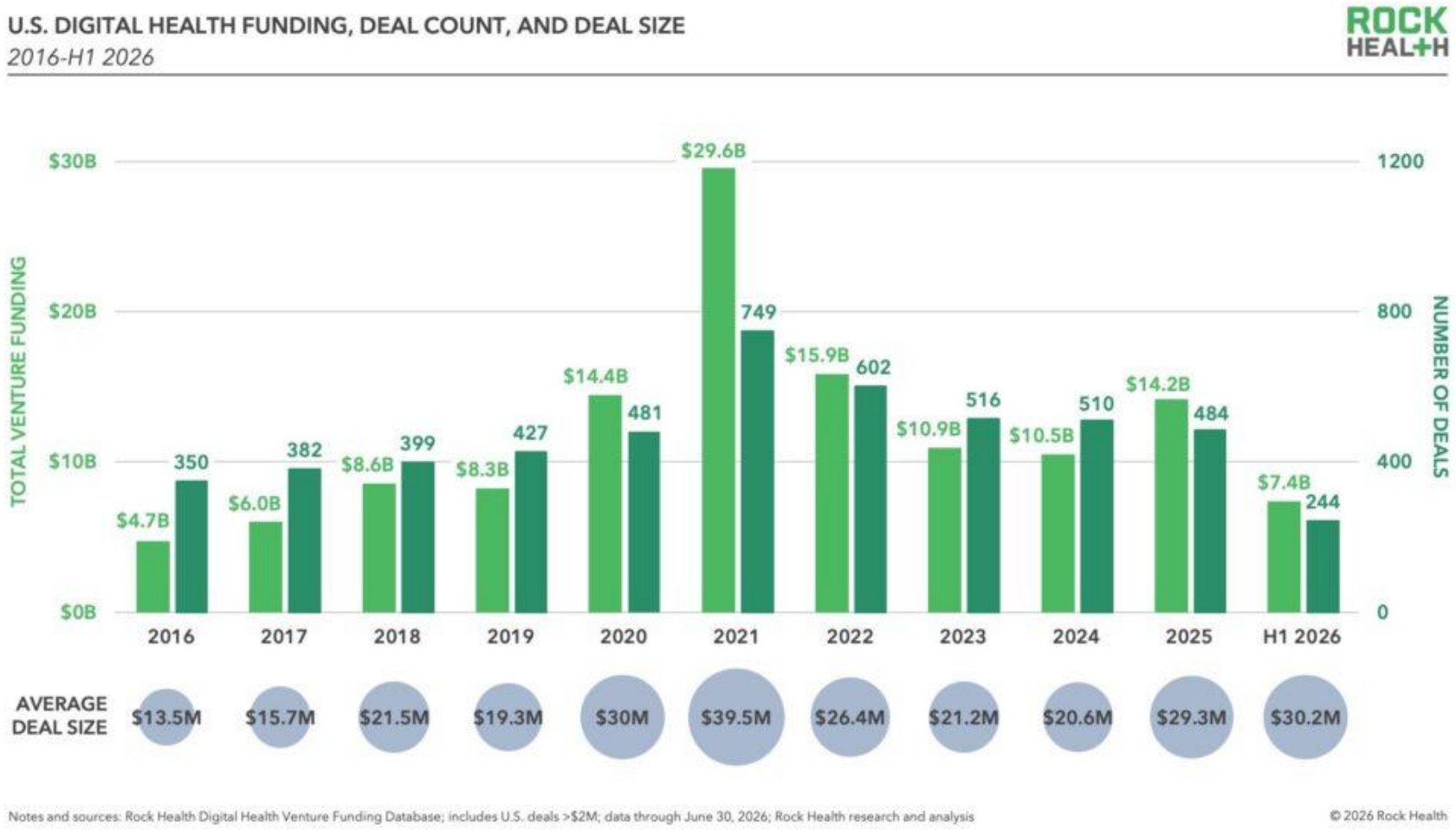




ChatGPT

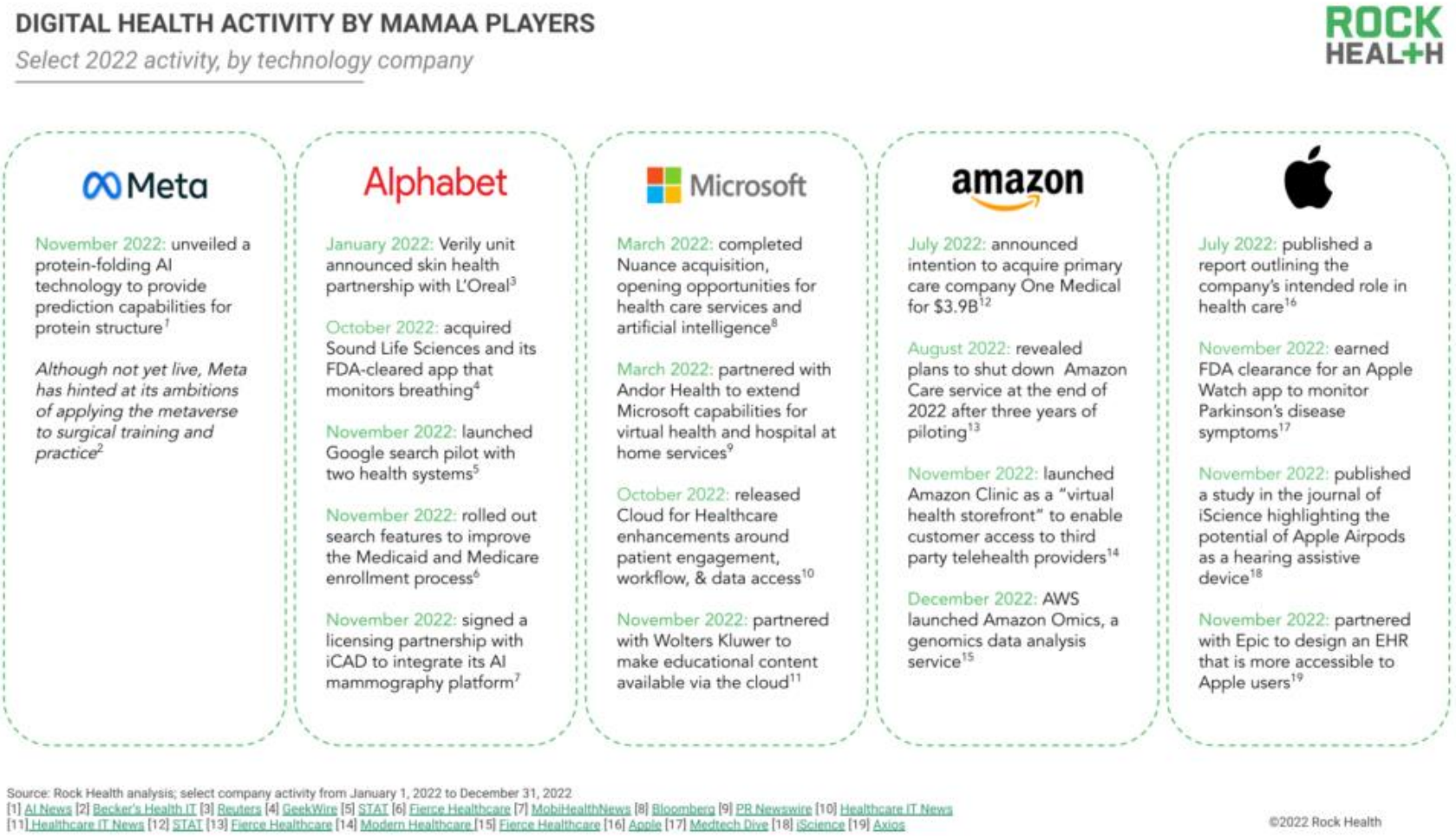
Massive Investments!

Digital Health Venture Funding



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Other Digital Health Investments



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Tech + Health Partnerships



2019

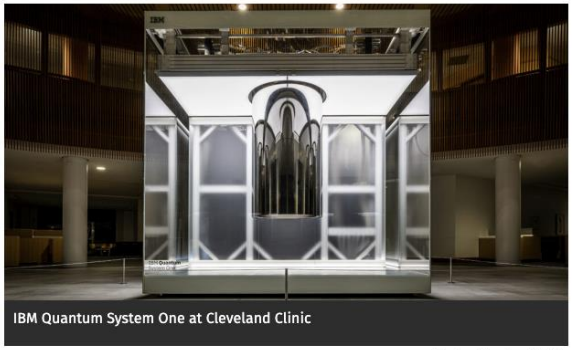
2023

Global Edition Artificial Intelligence

Cleveland Clinic, IBM to lead new quantum computing for health projects

Researchers will use $40 million in funding to develop quantum algorithms and workflows. They will also partner with Algorithmiq on a second initiative to create computational tools that demonstrate quantum advantage.

By Andrea Fox | October 25, 2023 | 02:39 PM



IBM Quantum System One at Cleveland Clinic

Photo: Cleveland Clinic

But Healthcare ML is not easy!

What makes healthcare different?

- Life or death decisions
  - Need robust algorithms
  - Checks and balances built into ML deployment

  (Also arises in other applications such as autonomous driving.)
  - Need for fairness and accountability

- Many questions are about unsupervised learning
  - Discovering disease subtypes.

- Many of the questions we want to answer are causal.
  - Naïve use of supervised machine learning is insufficient.

What makes healthcare different?

- Very little labeled data
  - Motivates semi-supervised or weakly-supervised learning algorithms.
- Sometimes small number of samples (e.g., a rare disease)
  - Learn as much as possible from other data (e.g., healthy subjects)
  - Model the problem carefully
- Lots of missing data, varying time intervals, censored labels

What makes healthcare different?

- Difficulty of de-identifying data
  - Need for data sharing agreements and sensitivity

- Difficulty of deploying ML
  - Commercial electronic health record software is difficult to modify.
  - Data are often in silos; everyone recognizes need for interoperability, but slow progress.
  - Careful testing and iteration are needed.

What makes healthcare different?

- Non-technical issues
  - Medicine is a highly-regulated field; need to comply with regulations.
  - Lack of understanding from clinicians, nurses, and other caretakers.
  - Lacking resources: GPUs and computers

Recent Regulations

U.S. FOOD & DRUG ADMINISTRATION

← Home / Medical Devices / Digital Health Center of Excellence / Software as a Medical Device (SaMD) / Artificial Intelligence in Software as a Medical Device

Artificial Intelligence in Software as a Medical Device



EU Artificial Intelligence Act

Some Deployment Experiences

Health AI Deployment Experiences

- 2016: Google published a deep learning model for diabetic retinopathy grading.


December 13, 2016

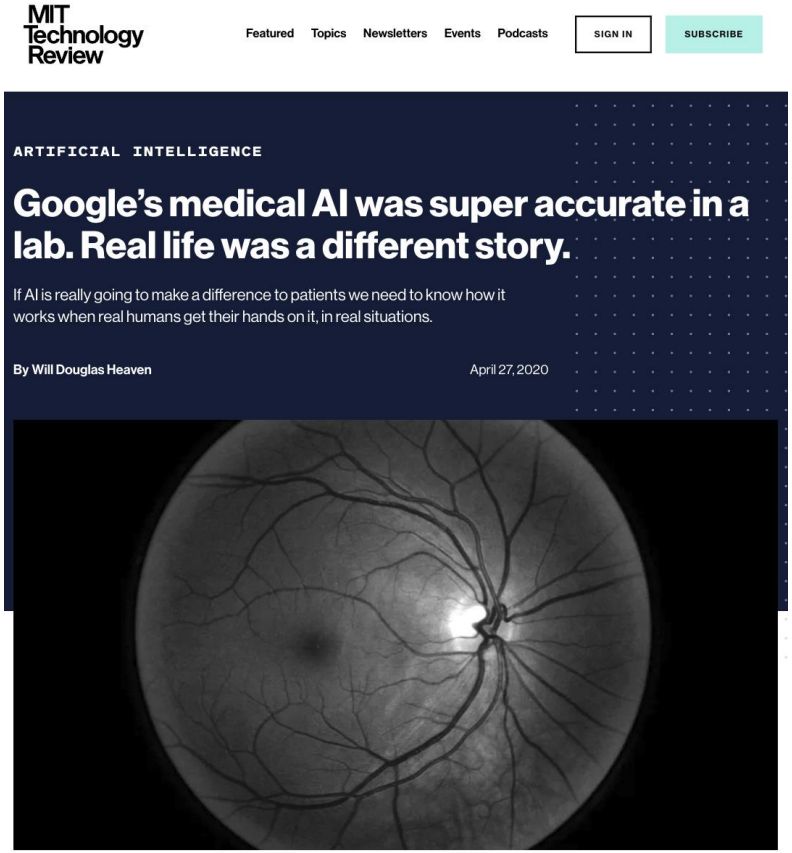
Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs

Varun Gulshan, PhD[1]; Lily Peng, MD, PhD[1]; Marc Coram, PhD[1]; et al

Author Affiliations | Article Information

JAMA. 2016;316(22):2402-2410. doi:10.1001/jama.2016.17216

- 2018 – 2019: Deployed in 11 hospitals in Thailand.
- Findings:
  - AI model struggled because of the different lighting conditions.
  - AI referred scans to specialists that were obviously normal to nurses.
  - It did speed up the process when it was right.
  - But in many cases, it caused unnecessary delays and increased workloads.


MIT Technology Review

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ARTIFICIAL INTELLIGENCE

Google's medical AI was super accurate in a lab. Real life was a different story.

If AI is really going to make a difference to patients we need to know how it works when real humans get their hands on it, in real situations.

By Will Douglas Heaven April 27, 2020

https://www.technologyreview.com/2020/04/27/1000658/google-medical-ai-accurate-lab-real-life-clinic-covid-diabetes-retina-disease/
https://dl.acm.org/doi/abs/10.1145/3313831.3376718

Health AI Deployment Experiences

Overview

IDx-DR

IDx-DR is a Class IIa Medical Device and the first ever FDA De Novo cleared autonomous AI.

2018



- IDx-DR referred 100% of patients with at least moderate diabetic retinopathy and more than 96% of patients with diabetic macular edema.
- IDx-DR avoided 91% of unnecessary specialty visits.
- IDx-DR has been trained and validated on over 2M images to mitigate bias and ensure the system works equally well for all people, regardless of sex, race, or ethnicity.

https://www.digitaldiagnostics.com/

Health AI Deployment Experiences

- Deep learning model for skin cancer detection – trained using 129,450 clinical images[1].



- Reported AUCs > 0.90, performing similar to expert dermatologists.

- An independent set of researchers validated the model on 656 expert curated, pathology confirmed images of diverse skin tones[2].

- Findings:
  - AUCs dropped by 27-36%.
  - AI model performed worse on darker skin tones and uncommon diseases.

[1] https://www.nature.com/articles/nature21056
[2] https://www.science.org/doi/10.1126/sciadv.abq6147

Health AI Deployment Experiences

- Epic's sepsis model was implemented in hundreds of hospitals to monitor patients and send an alert if they were at high risk for sepsis.
- The model uses a combination of real-time emergency room monitoring data (heart rate, blood pressure, etc.), demographic information, and information from the patient's medical records.
- An external validation by the U of Michigan resulted in very poor performance (AUC 0.63 vs the advertised AUCs of 0.73 and 0.83)[1].
- Out of 2,552 patients with sepsis, it only identified 33% of them and raised a lot of false alarms in the process.

[1]https://jamanetwork.com/journals/jamainternalmedicine/article-abstract/2781307

Health AI Deployment Experiences

- In 2016, IBM launched Watson Health to revolutionize healthcare with AI.
- It started with a high-profile partnership with Memorial Sloan Kettering to train an AI on EHR data to make treatment recommendations with investment of ~$5 billion.
- In 2022, IBM sold off all of Watson Health "for parts" for about $1 billion.

How IBM's Watson Went From the Future of Health Care to Sold Off for Parts

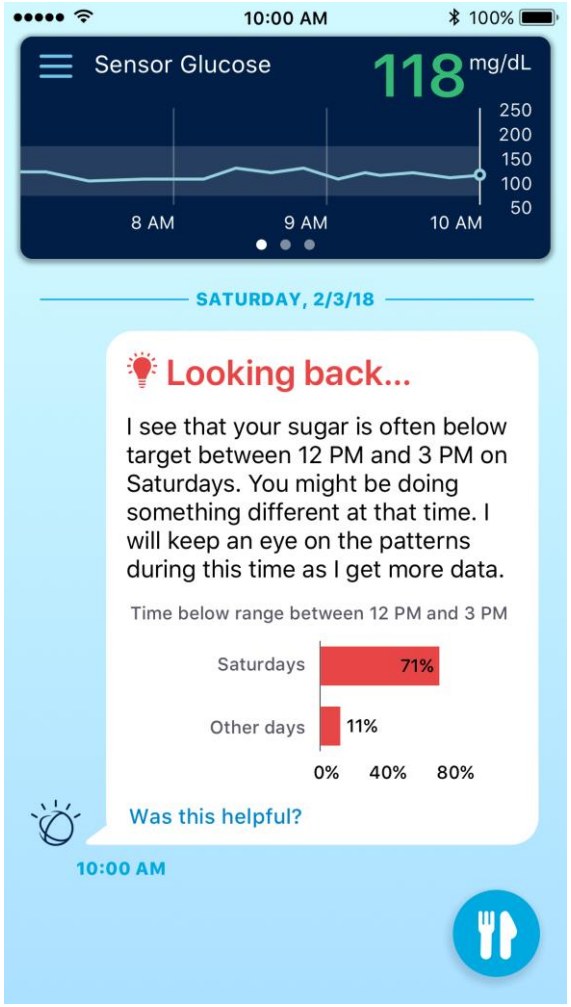
BY LIZZIE O'LEARY JAN 31, 2022 • 9:00 AM



https://slate.com/technology/2022/01/ibm-watson-health-failure-artificial-intelligence.html

Health AI Deployment Experiences

- Medtronic and IBM Watson Health leveraged their combined expertise in diabetes, AI and analytics to develop the Sugar.IQ diabetes assistant.
- These insights show how lifestyle choices, medications, and multiple daily injections impact diabetes management.
- The app uses AI and advanced analytics to give users a full picture of their current glucose levels and provides individualized guidance on daily diabetes management decisions.
- This system was approved by the FDA in 2018.



Discussion

- When does it work?

- When does it not work?

Questions?

https://z.umn.edu/ml4h_exitpass

